

Credit Default Prediction & Stress Testing Engine

Python | scikit-learn | Probability Calibration | Credit Risk | Stress Testing

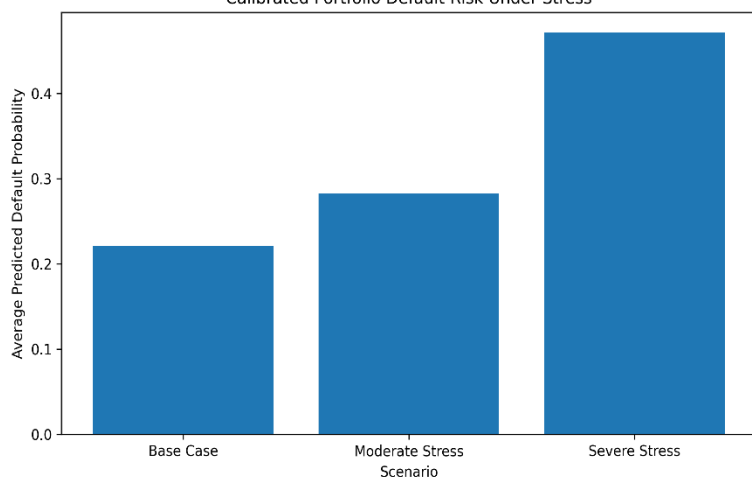
Executive Overview

Built a credit-risk modelling pipeline using 30,000 credit-card customers to predict default, calibrate probability estimates, segment borrowers by risk and simulate adverse repayment scenarios. Compared Logistic Regression and Random Forest models, then extended the analysis into probability calibration, borrower risk bands, behavioural stress testing and cost-sensitive threshold optimisation.

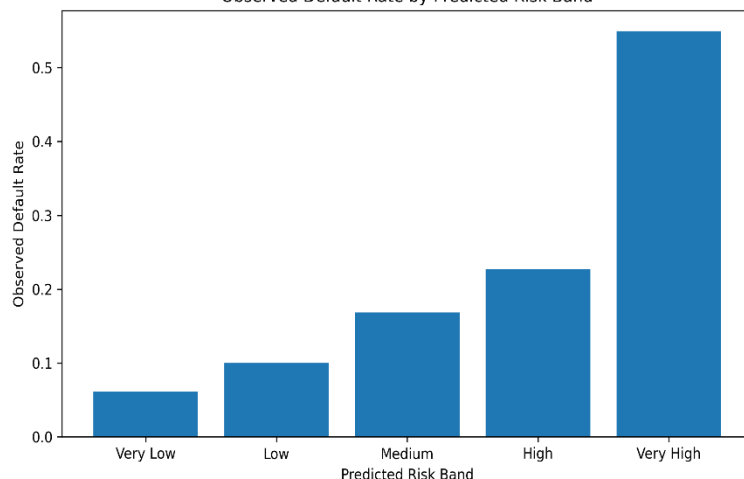
Key Results

- **Random Forest ROC-AUC:** 0.775 vs 0.754 for Logistic Regression
- **Brier score:** improved from 0.1696 to 0.1362 after calibration
- **Observed portfolio default rate:** 22.13%
- **Calibrated average predicted risk:** 22.08%
- **Highest-risk 20% of borrowers:** captured ~49.7% of all observed defaults
- **Very High-Risk default rate:** 54.96%, approximately 2.48× the portfolio average
- **Moderate stress:** predicted default risk increased to 28.31%
- **Severe stress:** predicted default risk increased to 47.17%

Calibrated Portfolio Default Risk Under Stress



Observed Default Rate by Predicted Risk Band



Analytical Conclusion

Repayment behaviour was the strongest driver of modelled default risk, particularly recent payment status, maximum delay and the number of delayed months. The Very High Risk segment also displayed materially lower payments and higher utilisation than the overall portfolio. Stress testing showed substantial deterioration in calibrated default risk under worsening borrower behaviour.

Under an illustrative 5:1 cost assumption for false negatives versus false positives, reducing the decision threshold from 0.50 to 0.15 increased recall from 38.3% to 74.4% and reduced the illustrative misclassification cost from 4,378 to 3,406 units, demonstrating why lending thresholds should reflect the economic cost of different errors.

Limitations

Dataset represents Taiwanese credit-card customers; stress scenarios are illustrative rather than regulatory forecasts; feature importance is predictive rather than causal; and threshold cost ratios are hypothetical rather than based on lender-specific loss data.